

Out-of-Sample Reweighting: Would a Realistic Demographic–Productivity Distribution Change the Productivity Effect of Blinding?

Fixed-behavior (composition) backbone — calibrated simulation, not an estimate

02 Jul 2026

1 Motivation and what this exercise can (and cannot) do

The experiment randomizes demographic labels onto fixed résumé content, so assigned demographics are **orthogonal to true productivity by construction**. Demographic bias can then only reshuffle candidates *within* a productivity level, never systematically select less-productive people — so the measured **null productivity effect of blinding is largely mechanical**. This note asks: if, as in the real market, demographic identity correlated with true productivity, would the measured biases translate into a productivity effect?

This is a **calibrated simulation, not an estimate**. It holds evaluator scoring behavior fixed (the “fixed-behavior” / composition channel) and sweeps the one thing the experiment cannot identify — the correlation between identity and the productivity that is *unobserved* by the score. The complementary statistical-discrimination channel (where blinding could destroy a genuinely predictive signal) is a bounded-invariance question deferred to a later layer (see §5).

2 Primitives (identified in-sample)

The content-score map $s(X)$ is fit on the 18 shown profiles’ blind-arm scores and predicted out to all 69; u is the true productivity **not** captured by that score. Firm prestige is now included in $s(X)$ (available for all 69).

Table 1: Content-score fit and the invisible-productivity share, by stage.

Stage	$s(X)$ LOO R2 (with firm)	LOO R2 (no firm)	SD(u)	invisible % var	tau (bias heterog.)
Engineer	0.90	0.91	7.85	14	0.62
HR	0.66	0.60	16.58	61	0.00

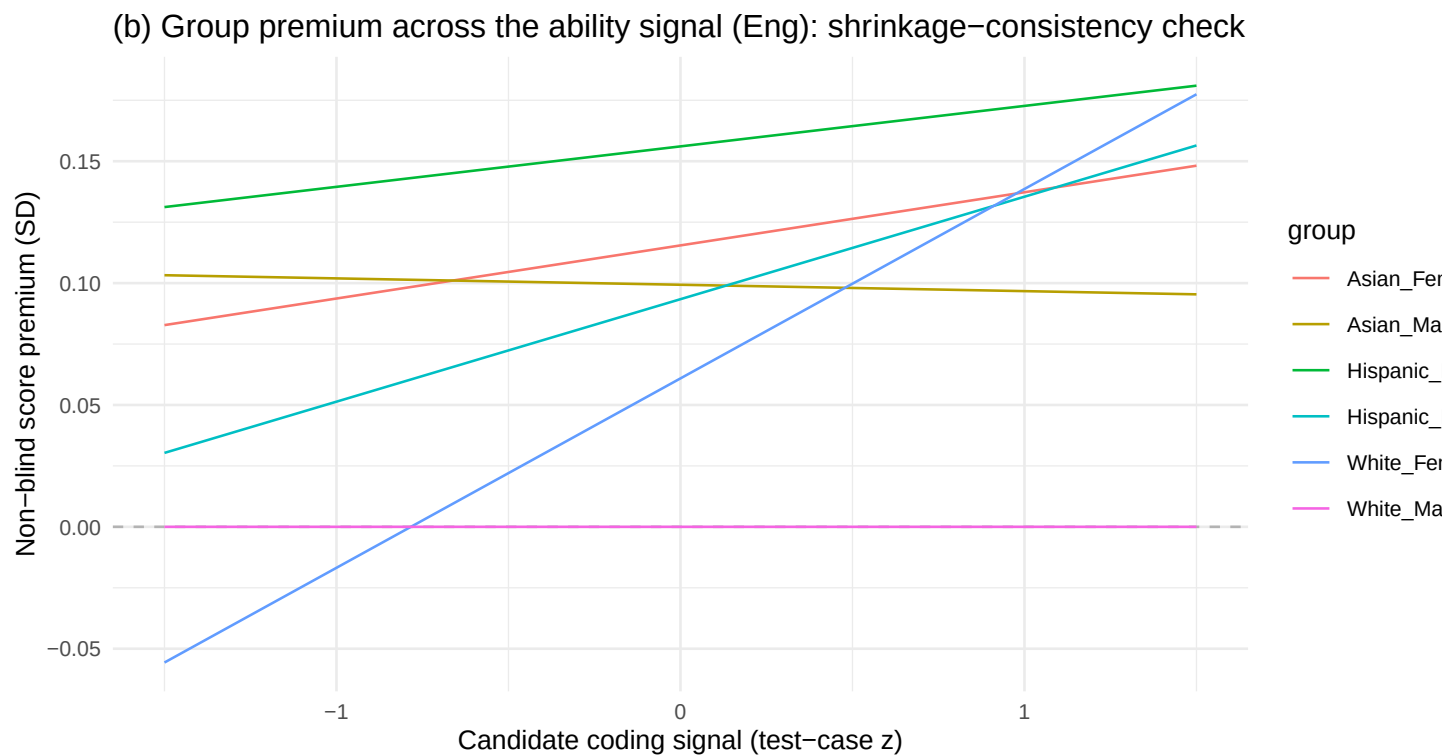
The score already explains the large majority of productivity variance, so only a small share (u) is left for identity to proxy. Adding firm prestige barely changes the fit (compare the two LOO columns) — consistent with prestige predicting the *score* more than lab productivity.

Table 2: Six-group non-blind score bias (demeaned) and pool share. Engineers show the S1.1 pattern (White Male penalized, minorities premium); HR biases are near zero. The largest-bias groups are the rarest in the pool.

Group	Eng bias	HR bias	pool share
White Male	-0.093	-0.032	0.10
White Female	-0.026	-0.013	0.03
Asian Male	0.007	-0.017	0.58
Asian Female	0.025	0.033	0.21
Hispanic Male	0.000	0.130	0.06
Hispanic Female	0.066	-0.079	0.01

3 (b) Does the bias bend with ability? A shrinkage-consistency check

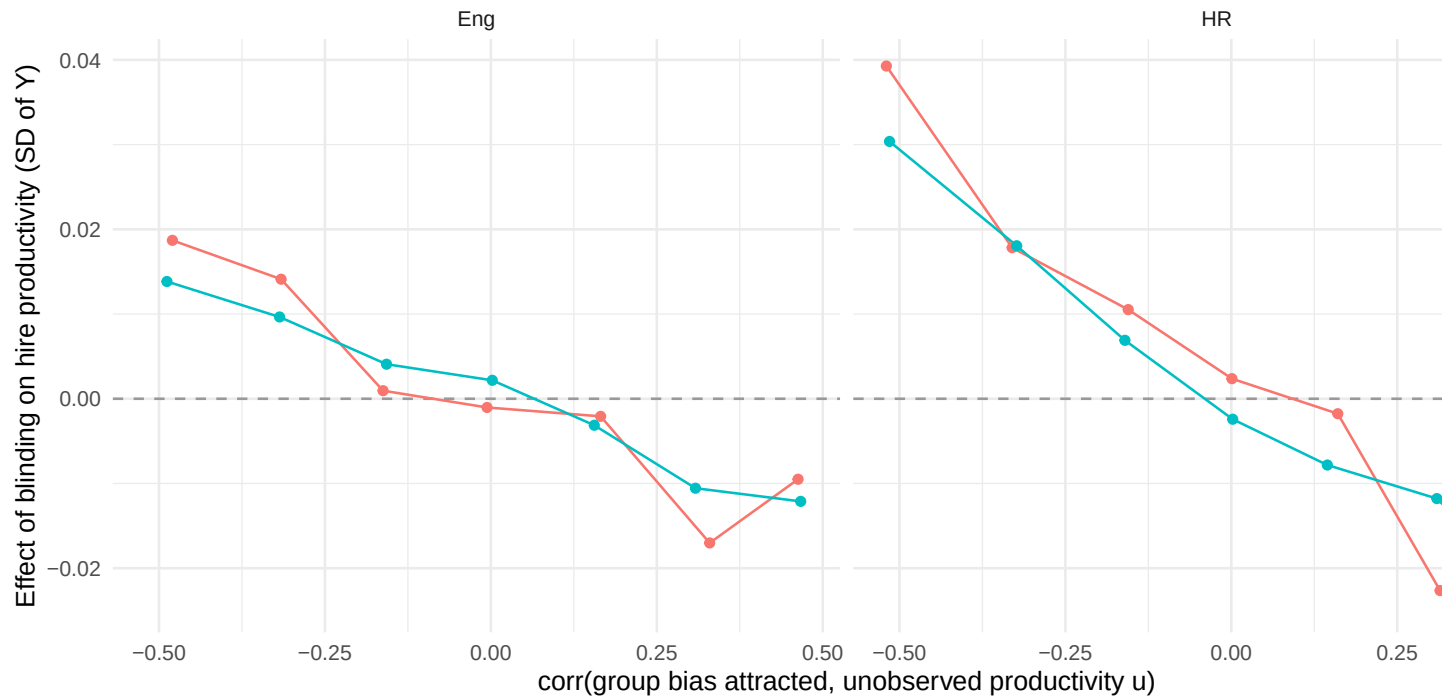
A pure level shift (consistent with constant taste) would be flat across the ability signal; a premium that *bends* toward zero as the signal strengthens is consistent with Bayesian shrinkage (statistical discrimination) — though, because taste can itself be ability-dependent, this is a **consistency check, not an identification** of the mechanism.



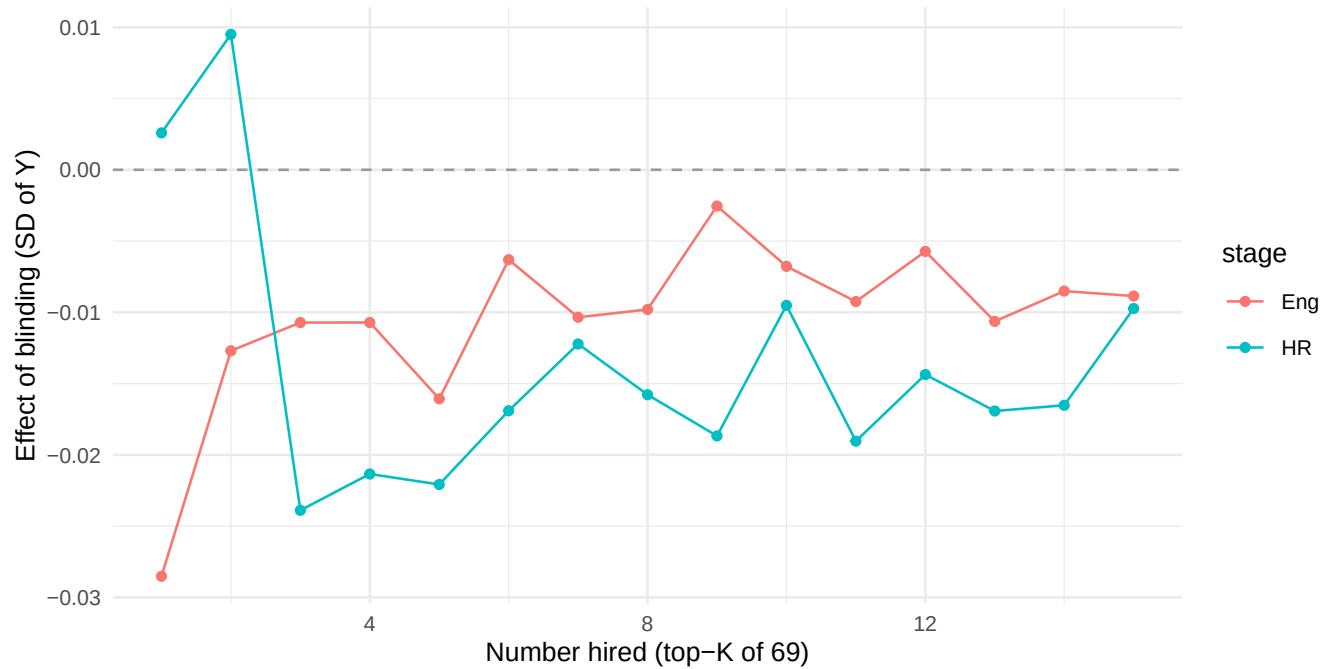
4 (f) Productivity effect of blinding: sensitivity and the hiring margin

Effect of blinding = $E[Y \mid \text{blind hires}] - E[Y \mid \text{non-blind hires}]$, in SD of true productivity. `corr = 0` recovers the measured null (validation).

(f) Sensitivity: productivity effect of blinding vs identity–productivity correlation



(f) k-curve at a strong anti-alignment (corr = -0.4)



Even at a **strong** identity–productivity correlation ($|\text{corr}|$ approx 0.5), the effect stays within roughly ± 0.02 SD for engineers and smaller for HR. The k-curve is flat and near zero across the hiring margin.

5 Sensitivity to the applicant pool (e)

Table 3: Blinding effect (SD of Y) at a strong anti-alignment ($\text{corr} = -0.4$, $K=10$): our Berkeley pool vs an illustrative US-representative pool.

stage	pool	blindeff
Eng	Our (Berkeley)	-0.005
Eng	US-representative	-0.017
HR	Our (Berkeley)	-0.014
HR	US-representative	-0.017

Reweighting to a US-representative demographic mix does not rescue a meaningful effect, because the mechanism is bounded by the biases and by $\text{SD}(u)$, not by the marginal shares alone.

6 Interpretation, the deferred layer, and caveats

Why the composition channel is inert here — three compounding reasons: (1) the biases are modest (~ 0.16 SD spread) and mostly cancel over a realistic pool; (2) visible signals already explain the large majority of productivity, leaving little u for identity to proxy; (3) the groups carrying the largest bias are the rarest in the pool. Refining the model (firm prestige, ability-tier-varying bias) sharpens but does not overturn this.

The real question is the deferred (c) layer. The composition channel cannot make blinding *hurt* efficiency; only a statistical-discrimination channel can (blinding removing a genuinely predictive signal). Our design cannot separate taste from statistical discrimination — and, since taste can be ability-dependent, not even the ability-slope cleanly identifies it. The honest object there is a productivity band swept over a **behavioral-invariance** parameter (scoring rule fixed $\rightarrow$ composition null; scoring rule re-optimizes to correct beliefs $\rightarrow$ efficiency cost). That band’s width *is* the statement of what the experiment cannot identify.

Caveats. Calibration, not estimation; $s(X)$ extrapolates from 18 profiles (LOO R^2 reported); group $\times$ tier interactions are low-powered at $N=18 \times 6$; the US-representative shares are illustrative; the Berkeley pool is 79% Asian with ~ 1 Hispanic-female profile.